LAB REPORT

LAB: CLUSTERING DATA

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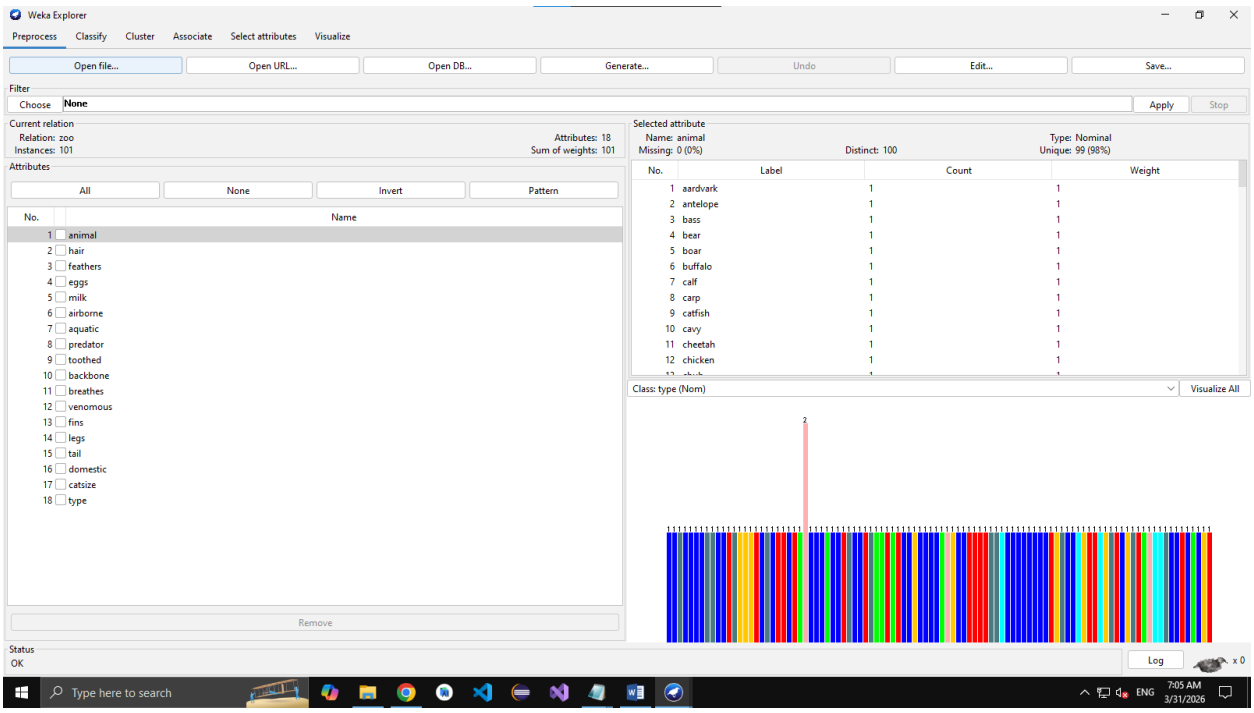
BÀI LÀM

Exercise 3: The goal of this data mining study is to find groups of animals in the zoo dataset, and to check whether these groups correspond to the real animal types in the dataset.



a. What types of variables are in this dataset?

Biến	animal	hair	feathers	eggs	milk	airborne	aquatic	predator	toothed
Kiểu DL	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal
Biến	backbone	breathes	venomous	fins	legs	tail	domestic	catsize	type
Kiểu DL	Nominal	Nominal	Nominal	Nominal	Numeric	Nominal	Nominal	Nominal	Nominal

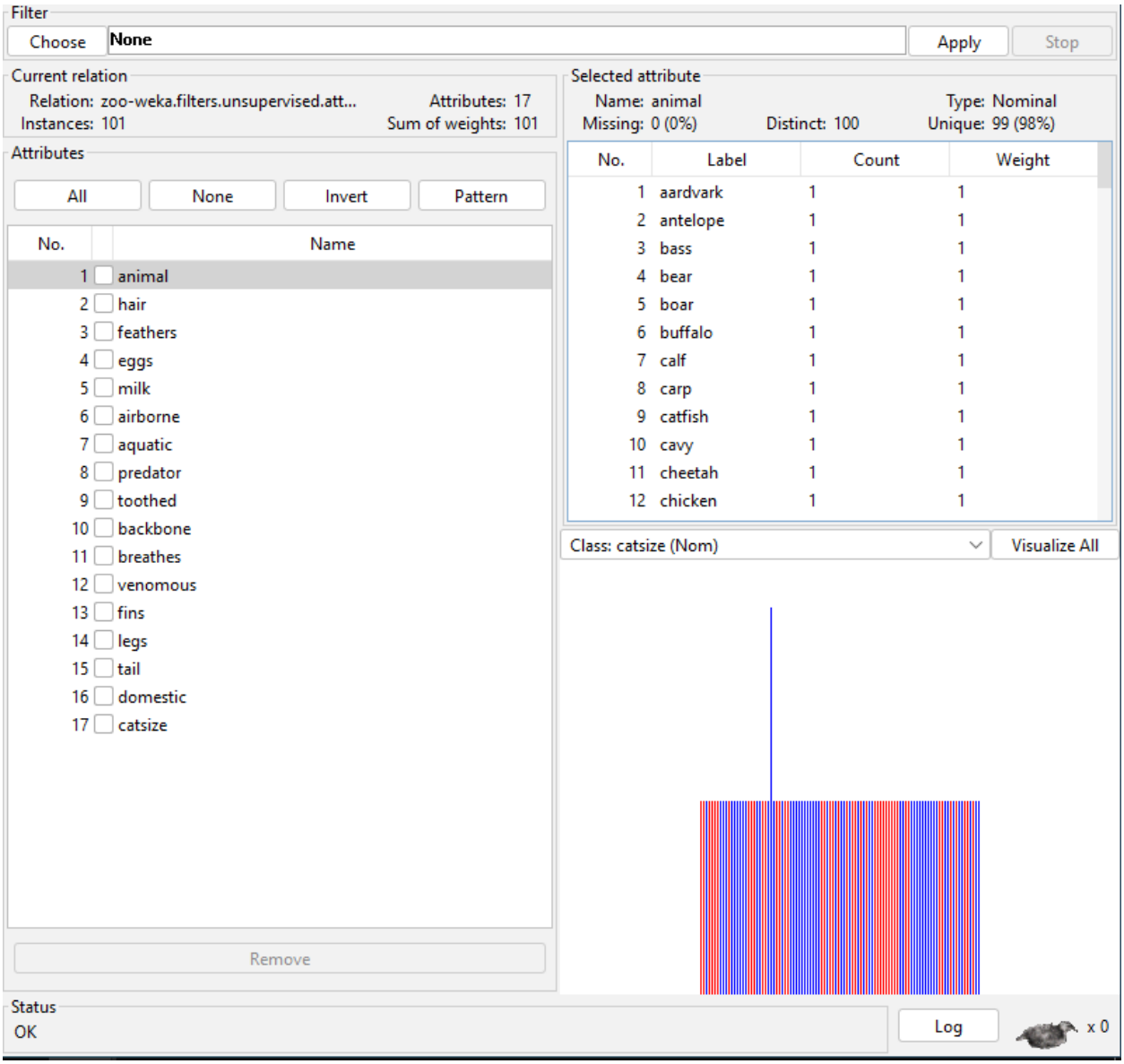
b. How many rows / cases are there?

→ Có 101 rows (instances)

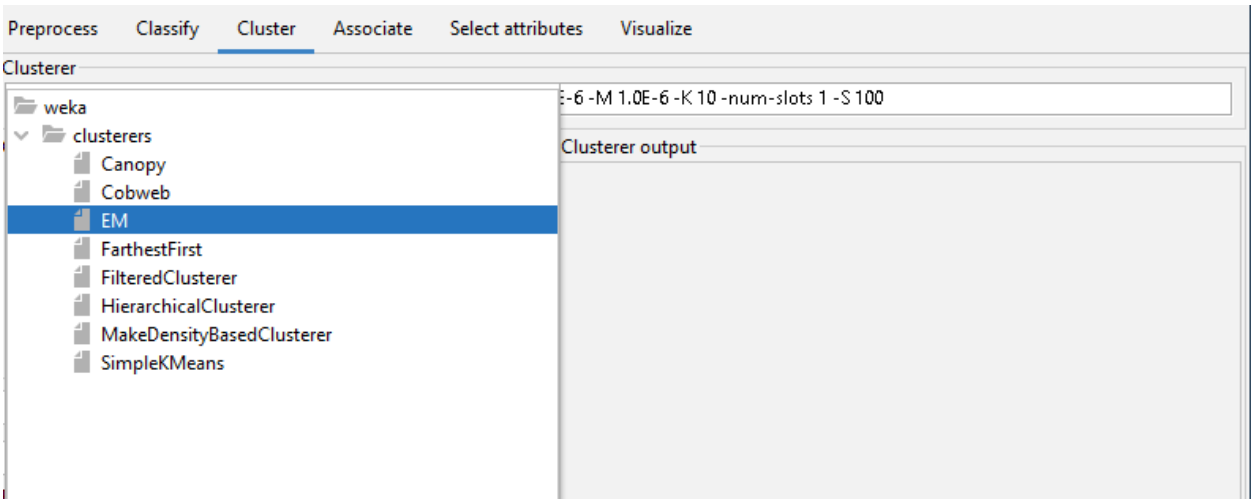
c. How many animal types are represented in this dataset? List them here.

→ 7 animal types: mammal, bird, reptile, fish, amphibian, insect, invertebrate

d. After removing the type attribute, go to the Cluster tab. How many clustering algorithms are available in Weka?



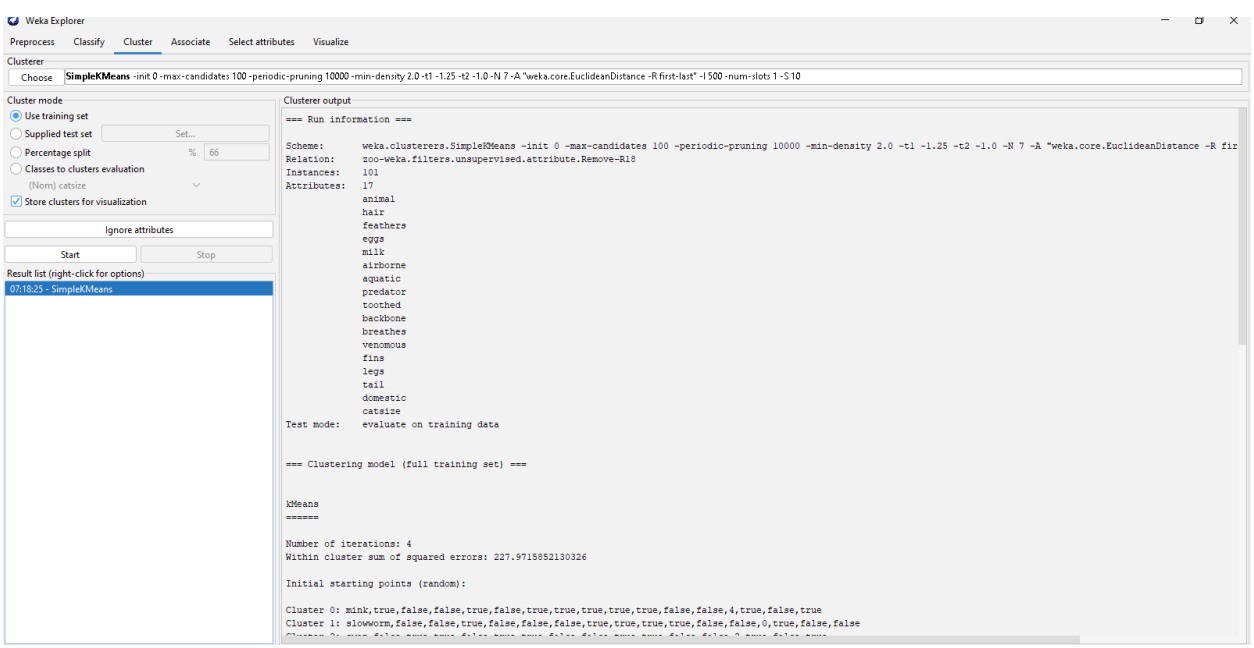
→ Có 5 thuật toán có thể sử dụng: SimpleKMeans, EM, FarthestFirst, HierarchicalClusterer, MakeDensityBasedClusterer



e. List the clustering algorithms seen in class, and map these to the ones provided in Weka.

Algorithm (Theory)	Weka Implementation
K-Means	SimpleKMeans
Hierarchical Clustering	HierarchicalClusterer
Expectation-Maximization	EM
Farthest-First Traversal	FarthestFirst
Density-Based (DBSCAN-like)	MakeDensityBasedClusterer

f. Start using the SimpleKMeans clusterer choosing 7 clusters. Do the clusters learnt and their centroids seem to match the animal types?



Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 7 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

- Use training set
- Supplied test set — Set...
- Percentage split — % 66
- Classes to clusters evaluation
  (Nom) catsize
- Store clusters for visualization

Ignore attributes

Start — Stop

Result list (right-click for options)

07:18:25 - SimpleKMeans

Clusterer output

```
Number of iterations: 4
Within cluster sum of squared errors: 227.9715852130326

Initial starting points (random):

Cluster 0: mink,true,false,false,true,false,true,true,true,true,true,false,false,4,true,false,true
Cluster 1: slowworm,false,false,true,false,false,false,true,true,true,true,false,false,0,true,false,false
Cluster 2: swan,false,true,true,false,true,true,false,false,true,true,false,false,2,true,false,true
Cluster 3: sole,false,false,true,false,false,true,false,true,true,false,false,true,0,true,false,false
Cluster 4: pitviper,false,false,true,false,false,false,true,true,true,true,true,false,0,true,false,false
Cluster 5: polecat,true,false,false,true,false,false,true,true,true,true,false,false,4,true,false,true
Cluster 6: parakeet,false,true,true,false,true,false,false,false,true,true,false,false,2,true,true,false

Missing values globally replaced with mean/mode

Final cluster centroids:
                           Cluster#
Attribute    Full Data          0          1          2          3          4          5          6
               (101.0)      (6.0)     (19.0)     (11.0)     (14.0)      (7.0)     (35.0)      (9.0)
==========================================================================================================
animal           frog    dolphin       clam       crow       bass       frog   aardvark    chicken
hair            false       true      false      false      false      false       true      false
feathers        false      false      false       true      false      false      false       true
eggs             true      false       true       true       true       true      false       true
milk            false       true      false      false      false      false       true      false
airborne        false      false      false       true      false      false      false       true
aquatic         false       true      false       true       true       true      false      false
predator         true       true      false       true       true       true      false      false
toothed          true       true      false      false       true       true       true      false
backbone         true       true      false       true       true       true       true       true
breathes         true       true       true       true      false       true       true       true
venomous        false      false      false      false      false      false      false      false
fins            false       true      false      false       true      false      false      false
legs           2.8416     1.6667     4.6842          2          0     2.8571     3.6571          2
tail             true       true      false       true       true       true       true       true
domestic        false      false      false      false      false      false      false      false
catsize         false       true      false      false      false      false       true      false
```

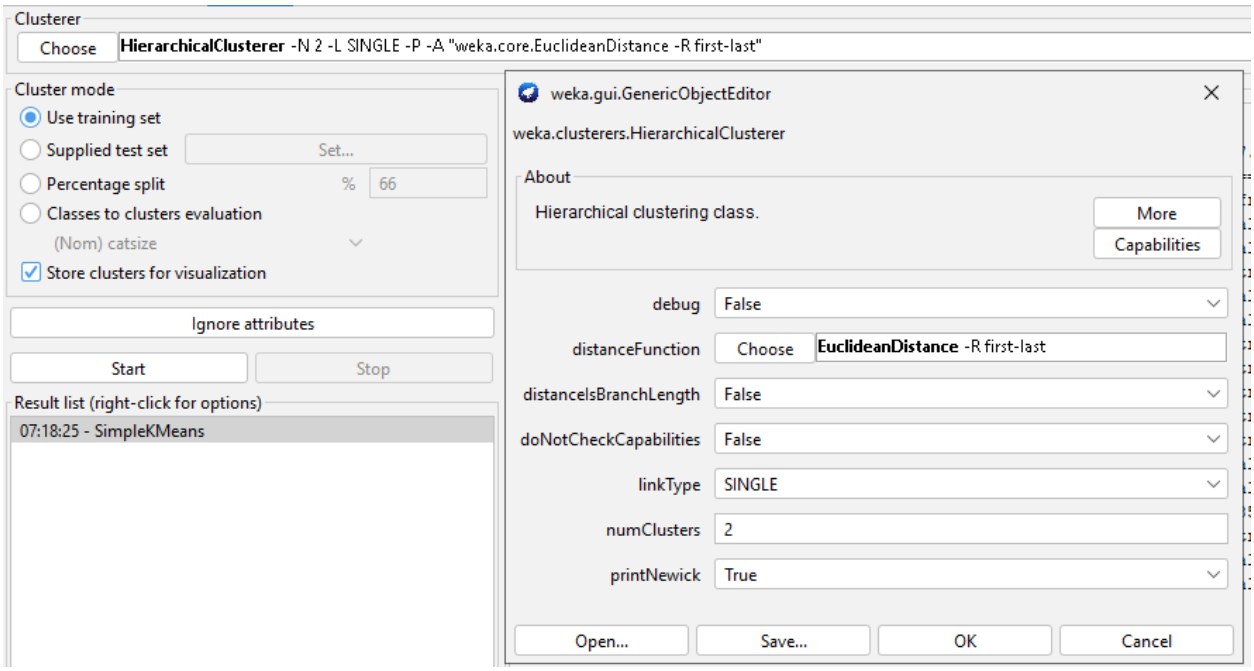
```
Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0         6 (  6%)
1        19 ( 19%)
2        11 ( 11%)
3        14 ( 14%)
4         7 (  7%)
5        35 ( 35%)
6         9 (  9%)
```

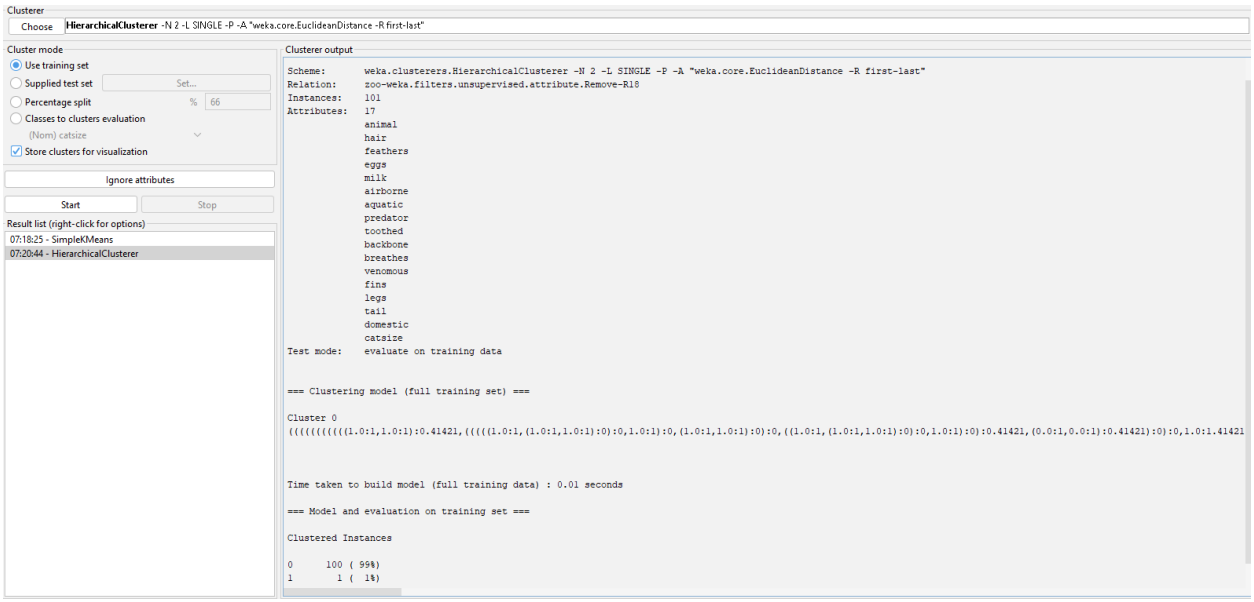
→ Các cụm và centroid thu được từ SimpleKMeans với k = 7 chỉ phản ánh một phần các loại động vật trong dữ liệu. Một số nhóm như mammals, birds và fish được nhận diện tốt, nhưng các nhóm khác bị chồng lấn. Do đó, kết quả phân cụm không hoàn toàn tương ứng với phân loại thực tế.

- Một số cụm có centroid rất rõ ràng và phản ánh đúng loại động vật, ví dụ:
  + Cụm có đặc trưng hair = 1, milk = 1 → tương ứng với mammals
  + Cụm có feathers = 1, eggs = 1 → tương ứng với birds
  + Cụm có aquatic = 1, fins = 1 → tương ứng với fish

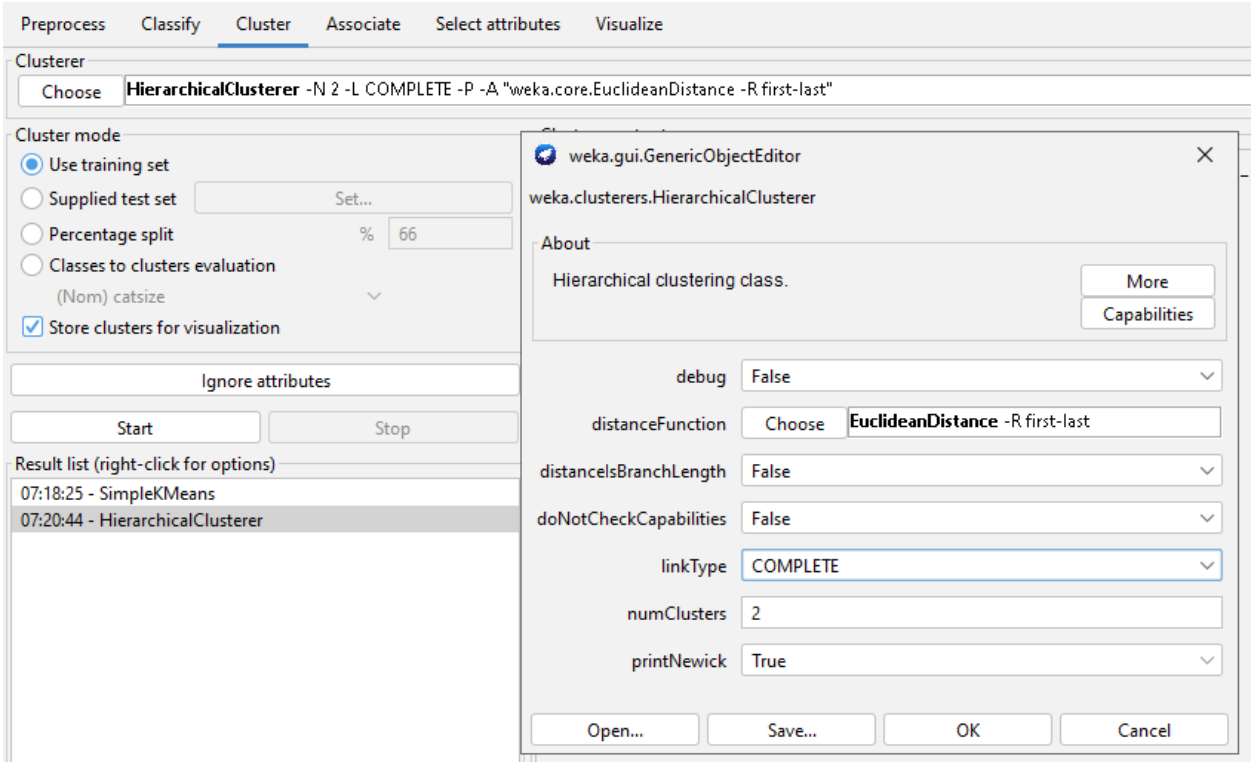
- Tuy nhiên, một số cụm không phân biệt rõ ràng, ví dụ: Reptiles và Amphibians có nhiều đặc điểm giống nhau nên dễ bị gộp chung; Insects và Invertebrates cũng có thể bị trộn lẫn

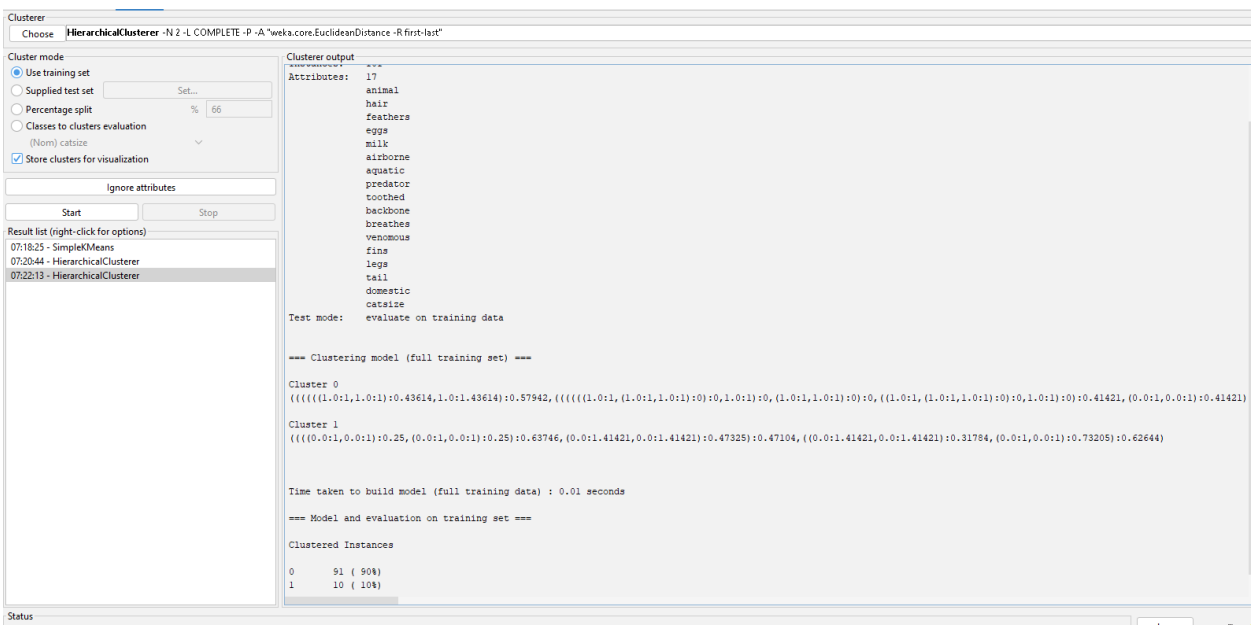
g. Start using the HierarchicalClusterer with Single Link.





h. Start using the HierarchicalClusterer with Complete Link.





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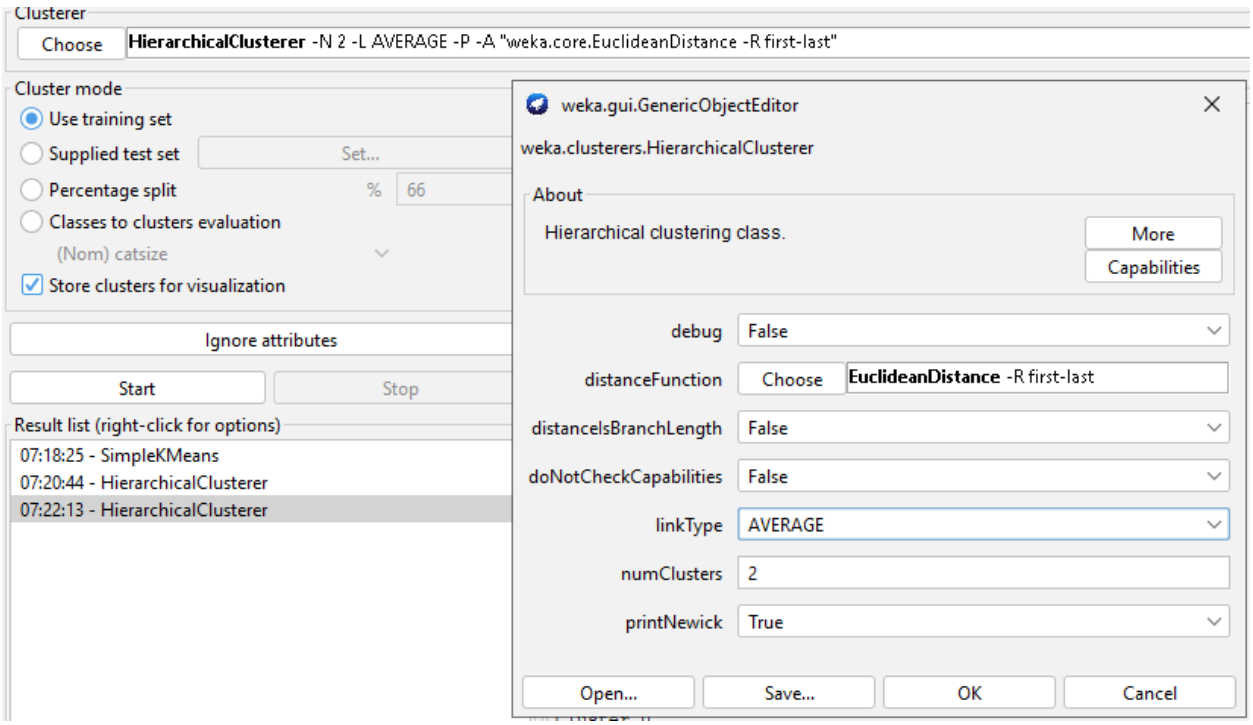
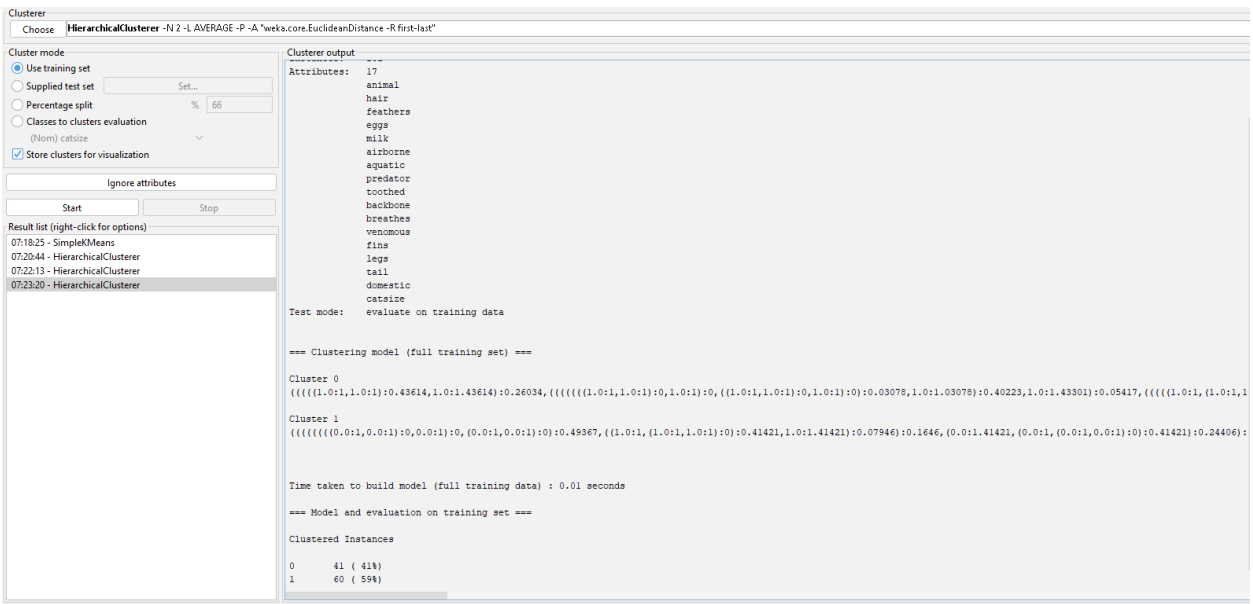
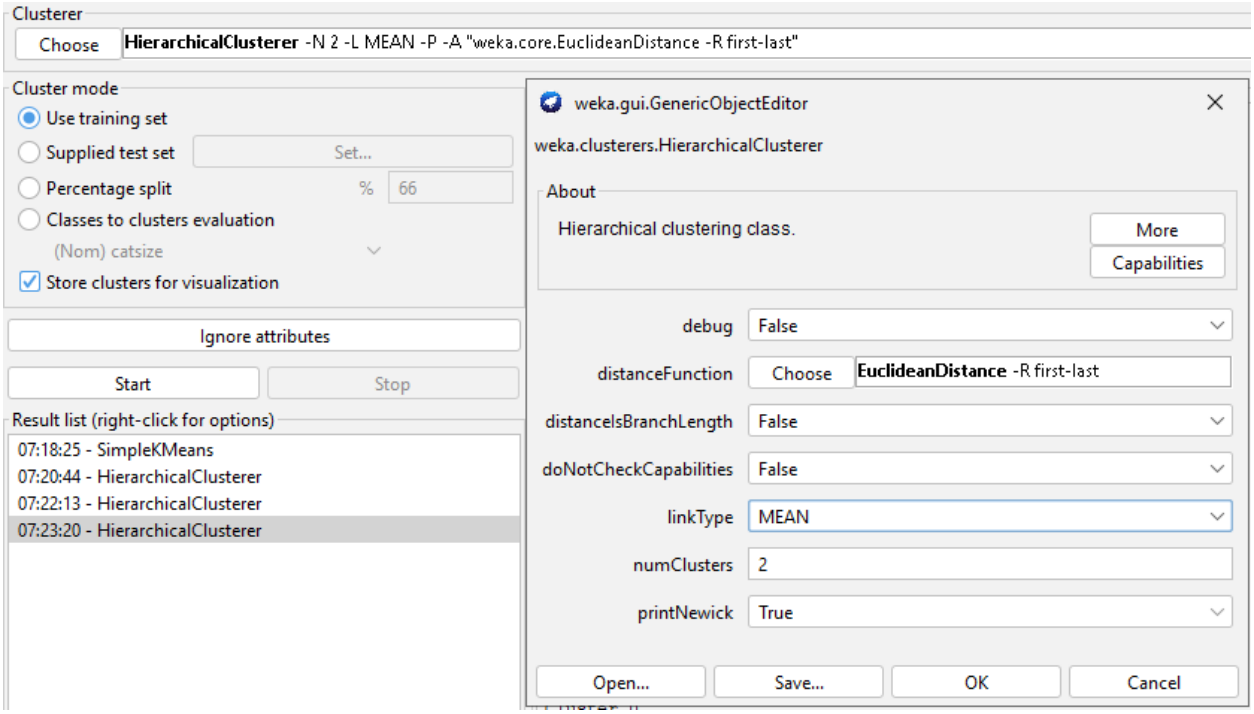
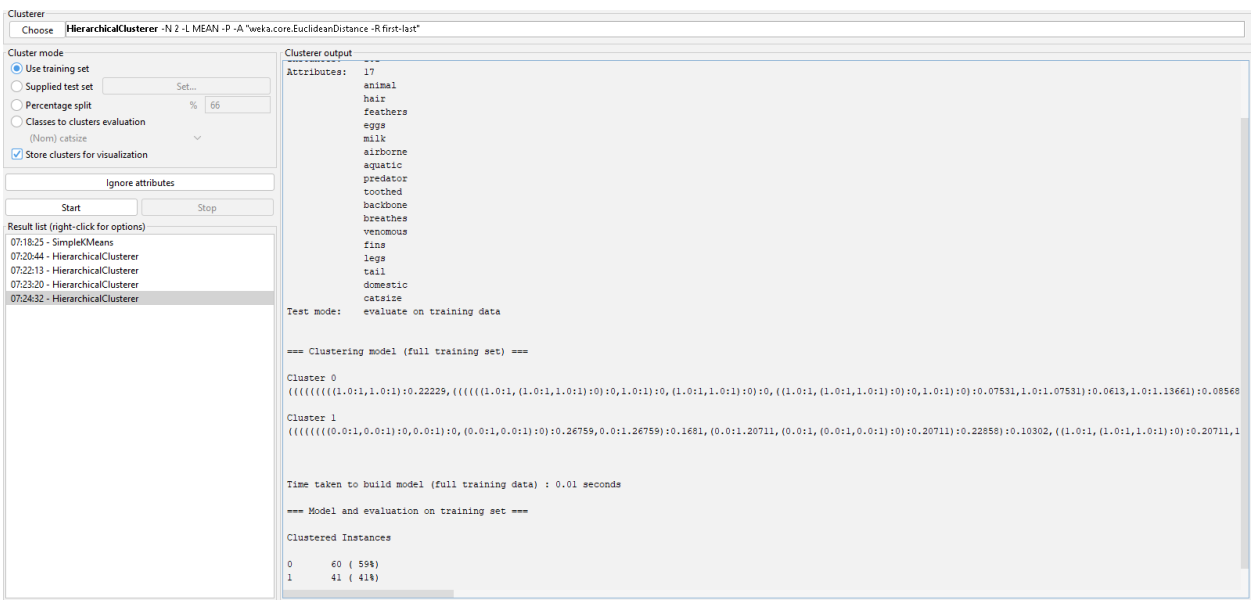


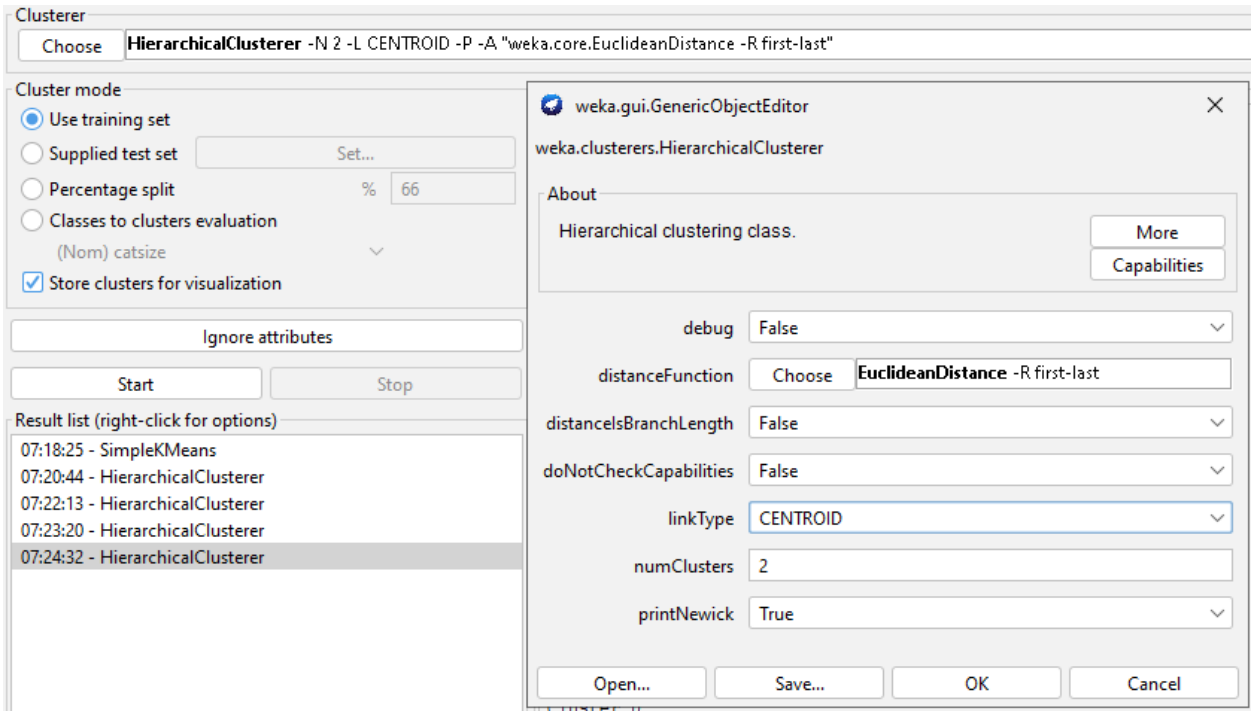


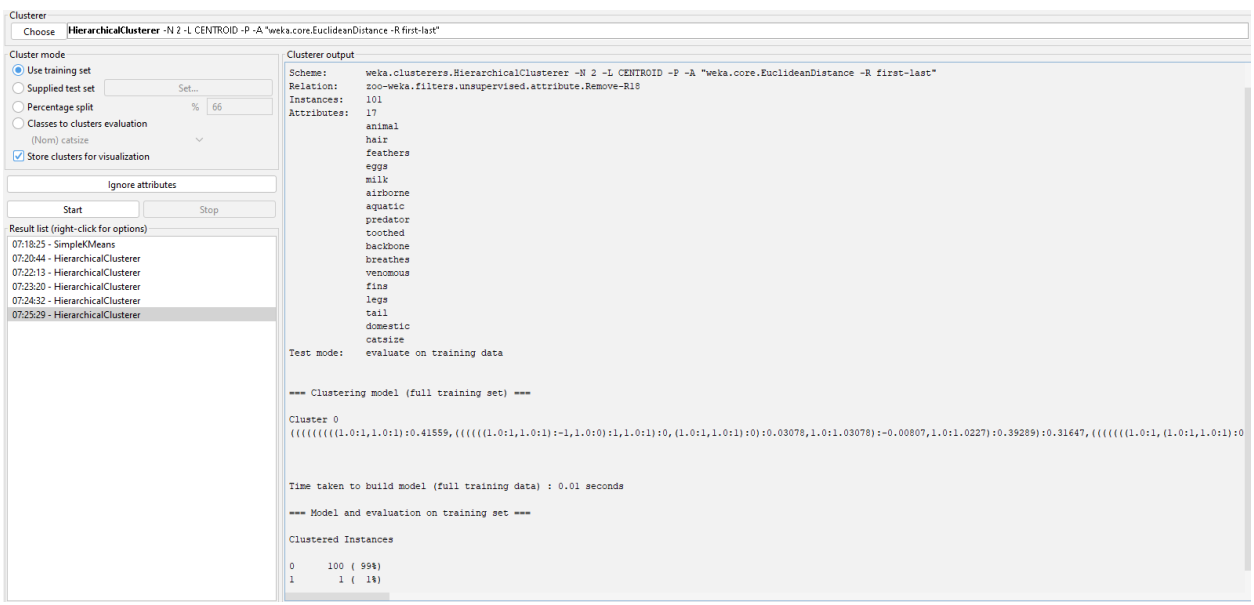
j. Start using the HierarchicalClusterer with Mean Link.



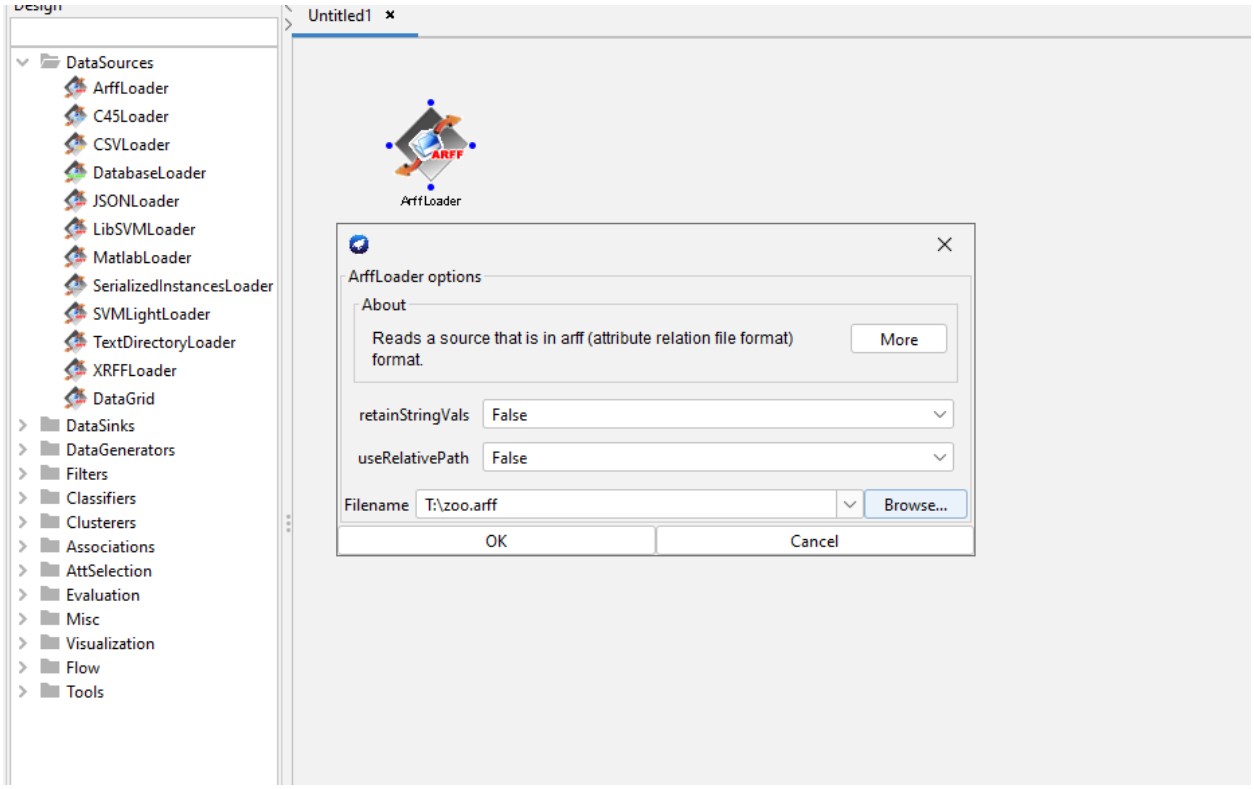


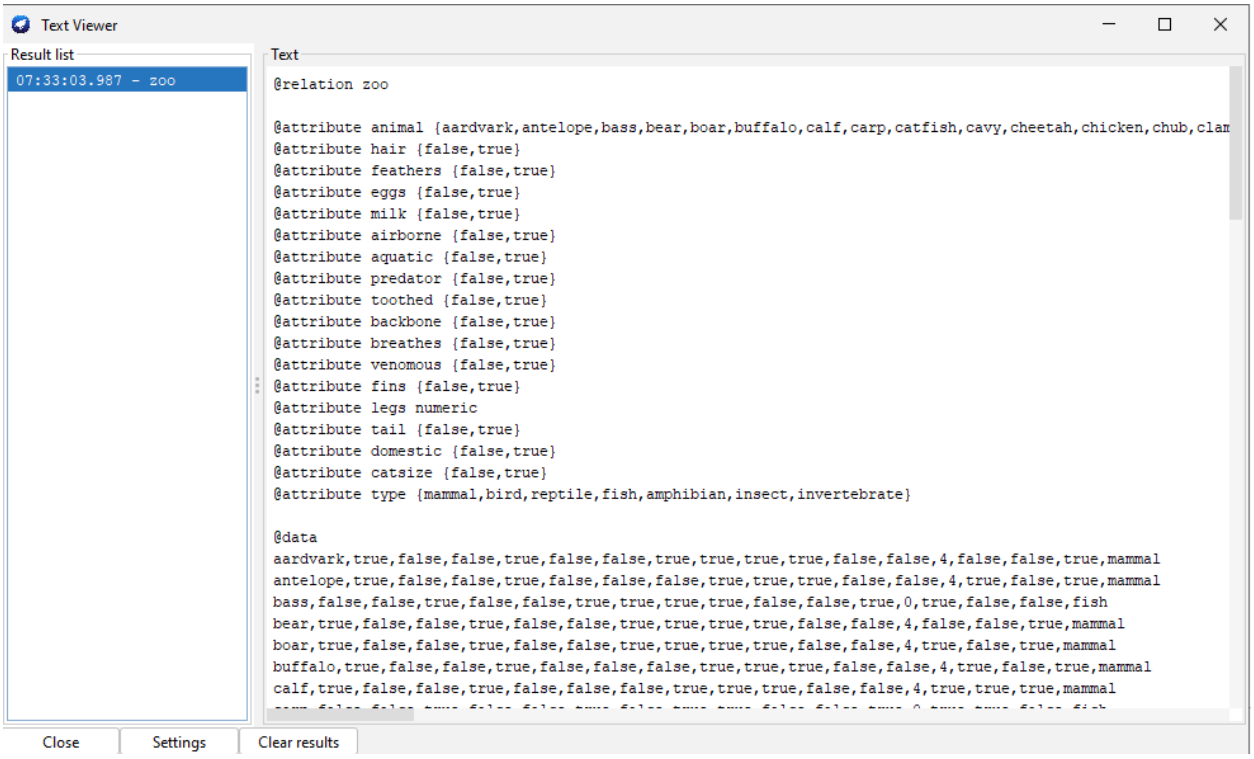
k. Start using the HierarchicalClusterer with Centroid Link.





Exercise 4: Using Weka Knowledge Flow to cluster data for the above exercises.





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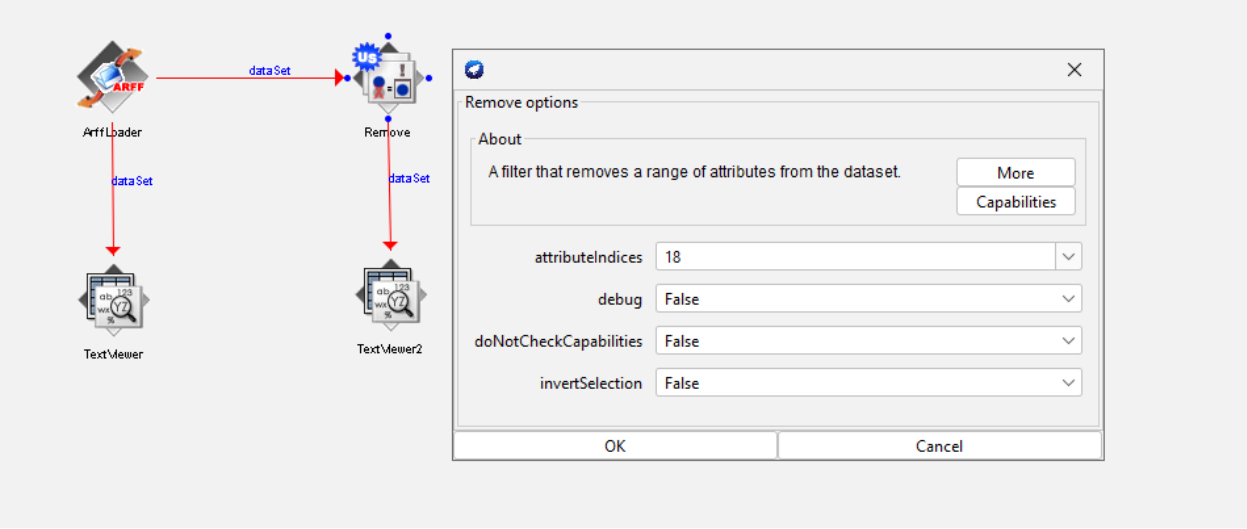
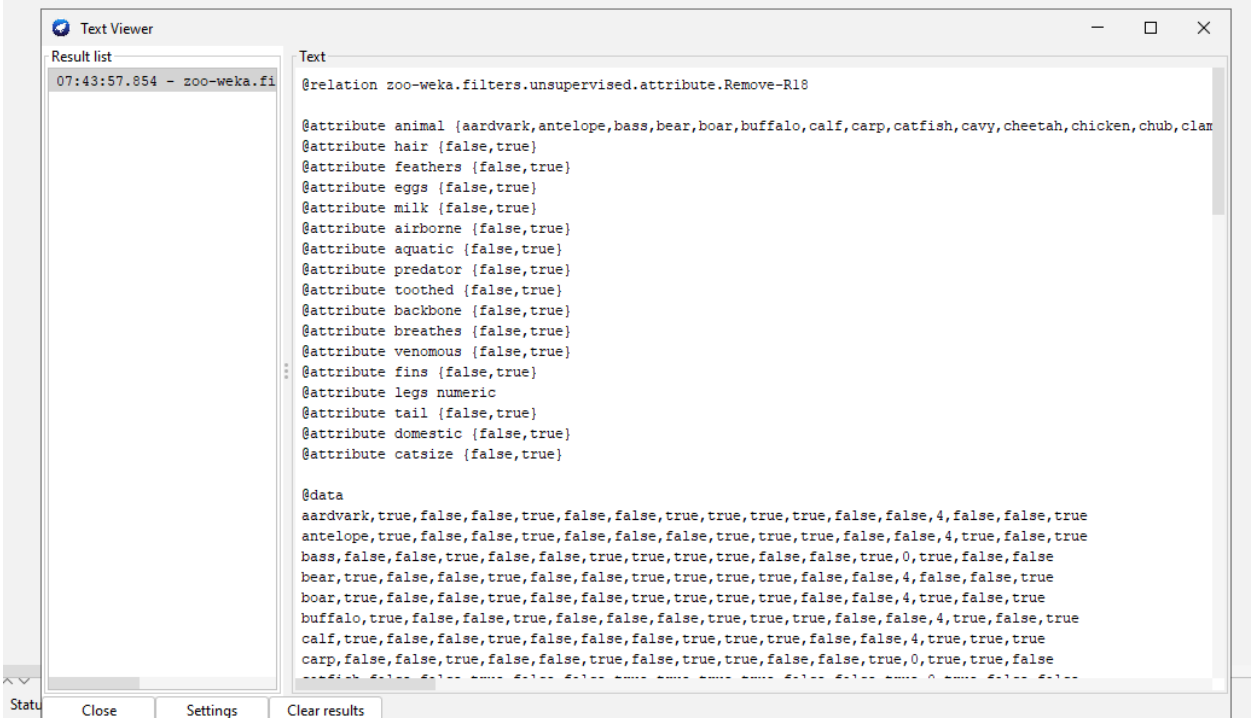
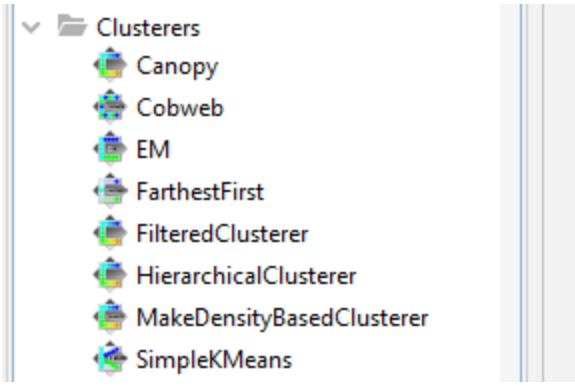
→ 7 animal types: mammal, bird, reptile, fish, amphibian, insect, invertebrate

d. After removing the type attribute, go to the Cluster tab. How many clustering algorithms are available in Weka?

→ Trong Filters: unsupervised - attribute, chọn Remove và set AttributeIndices = 18 (Index của type)







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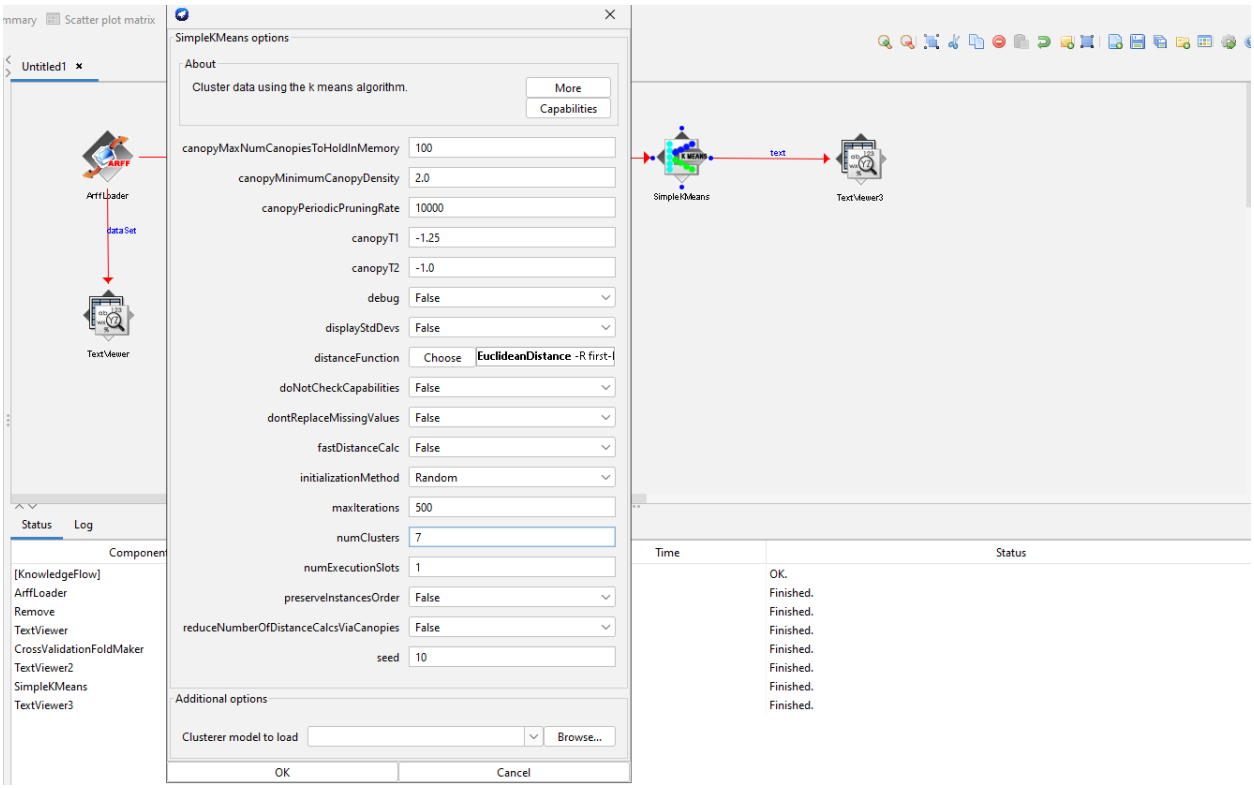

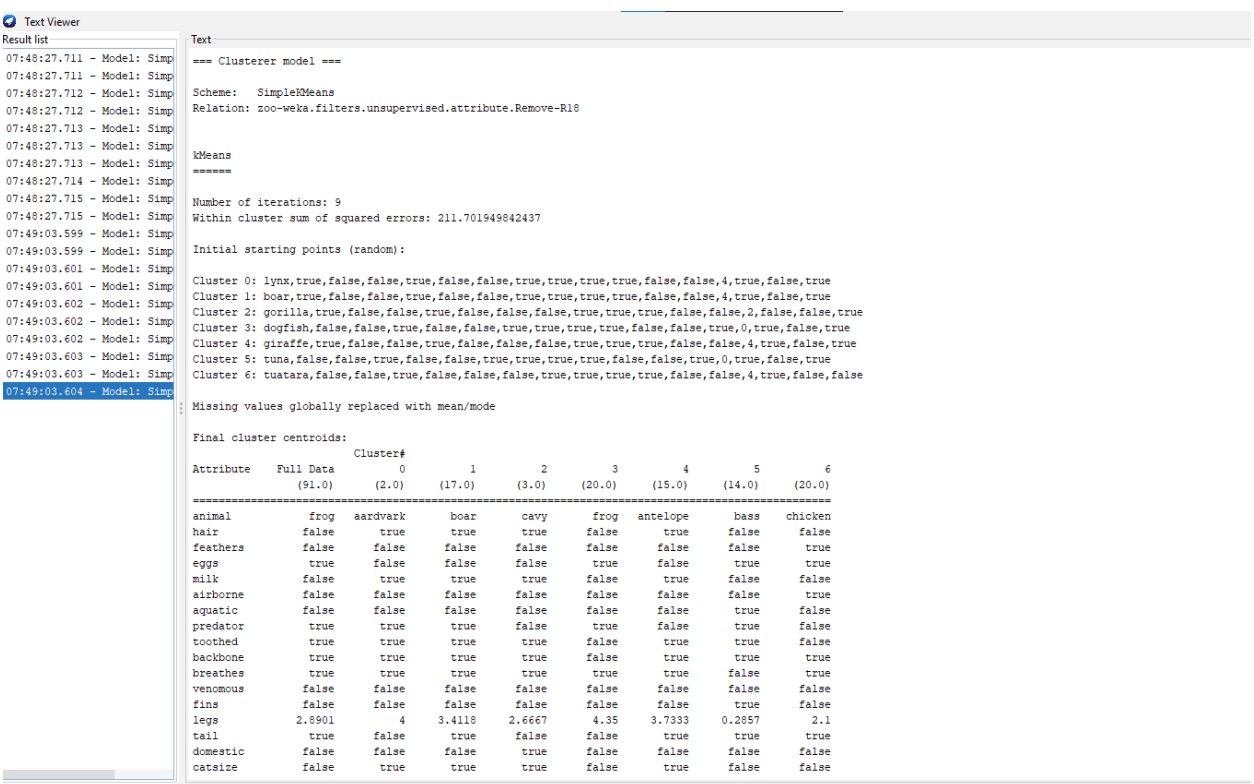

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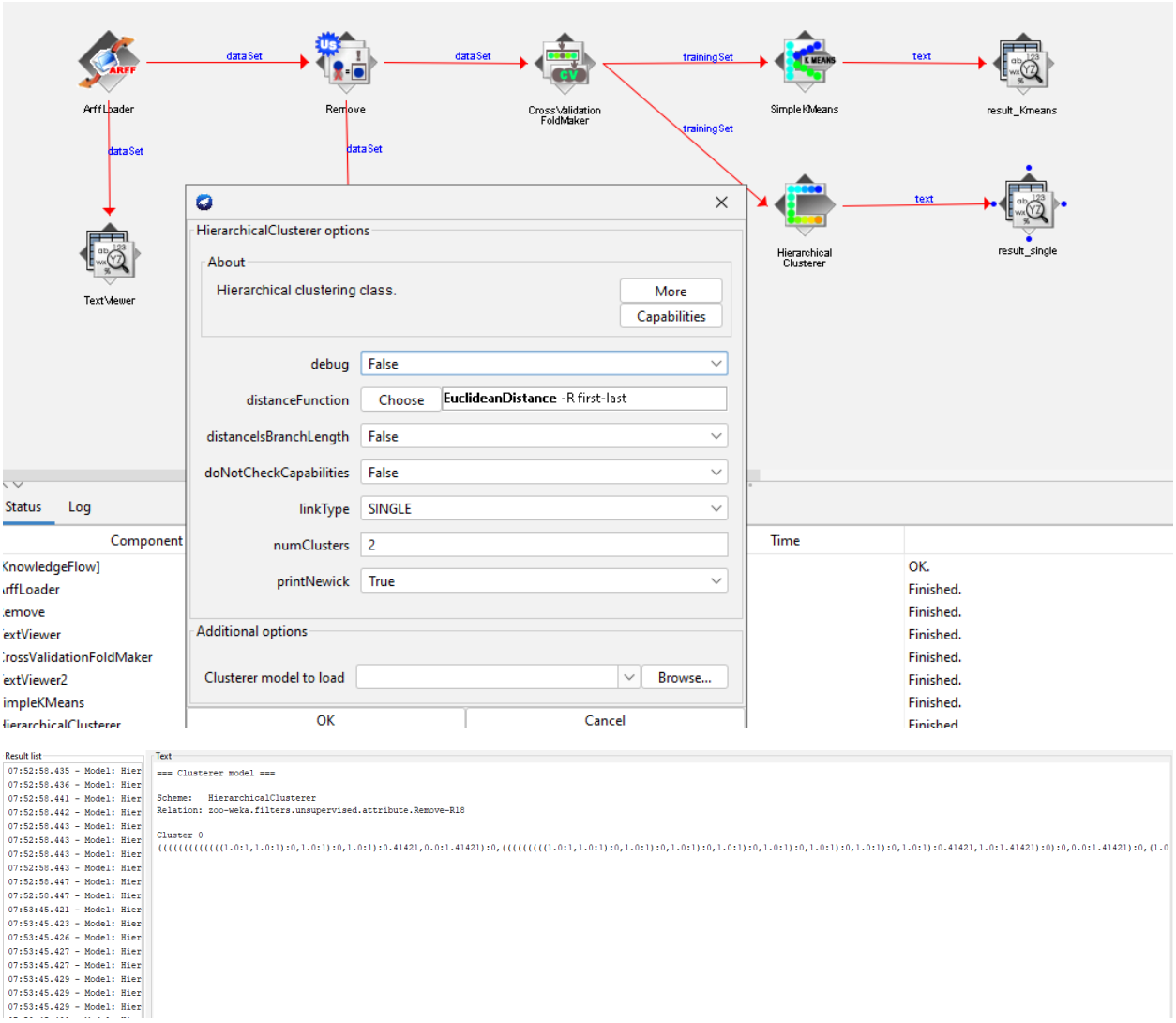

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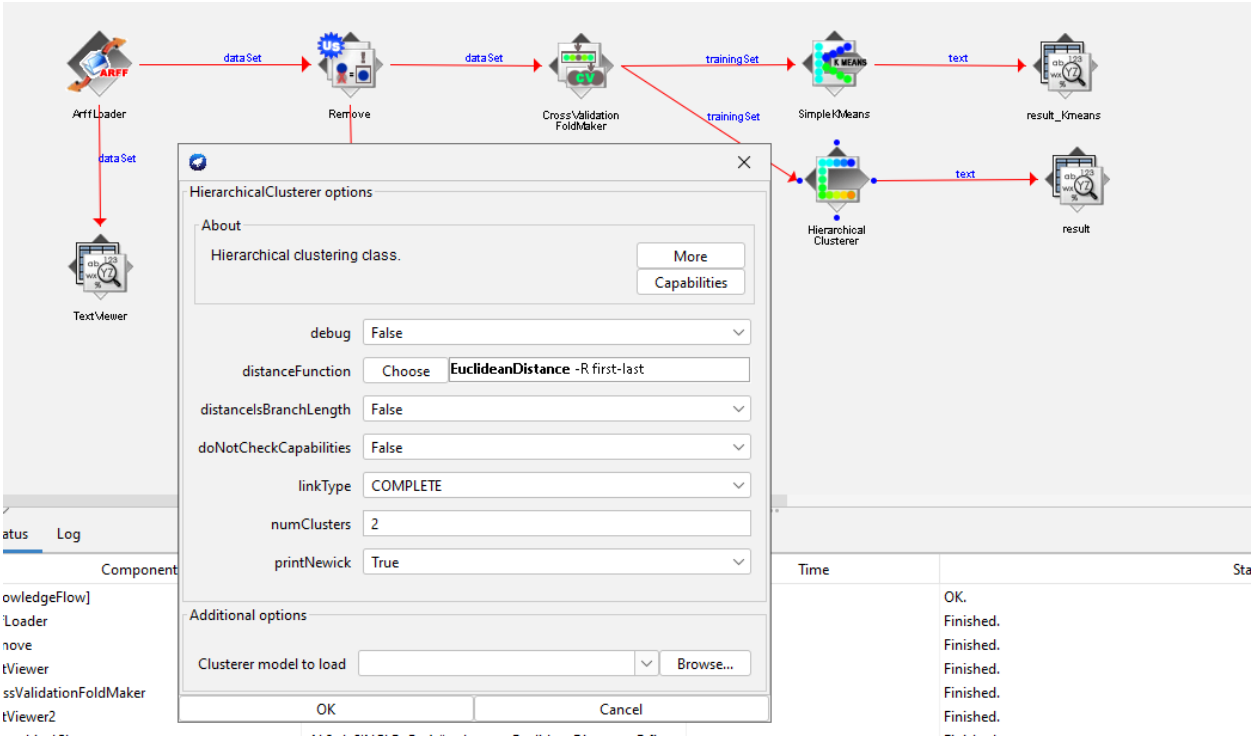

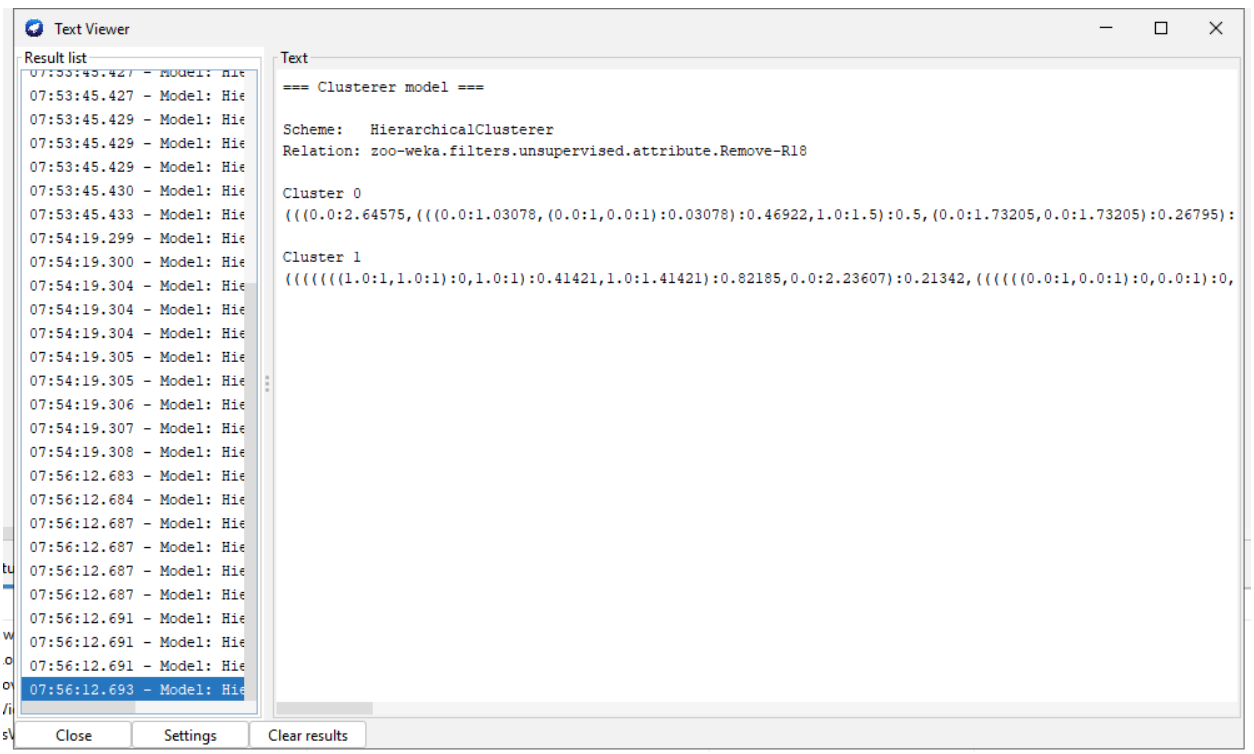

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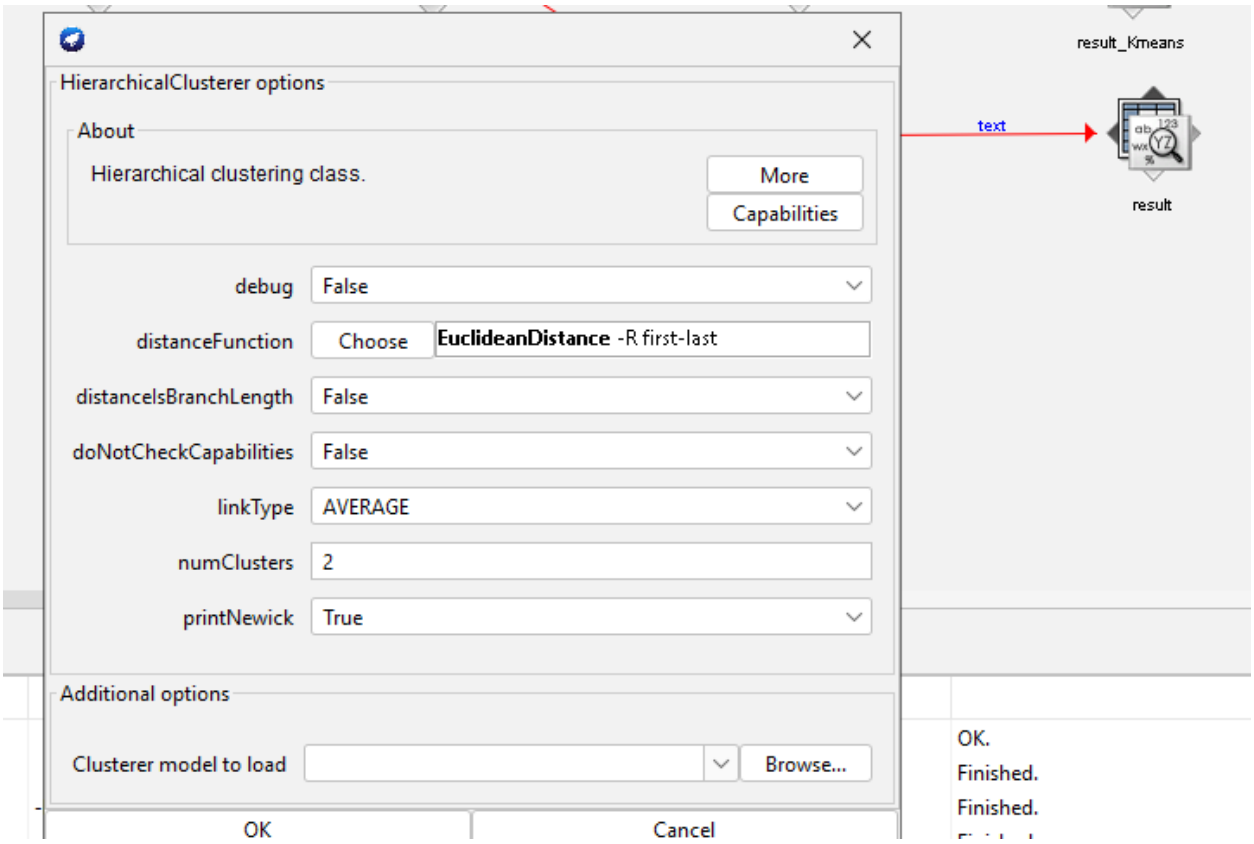
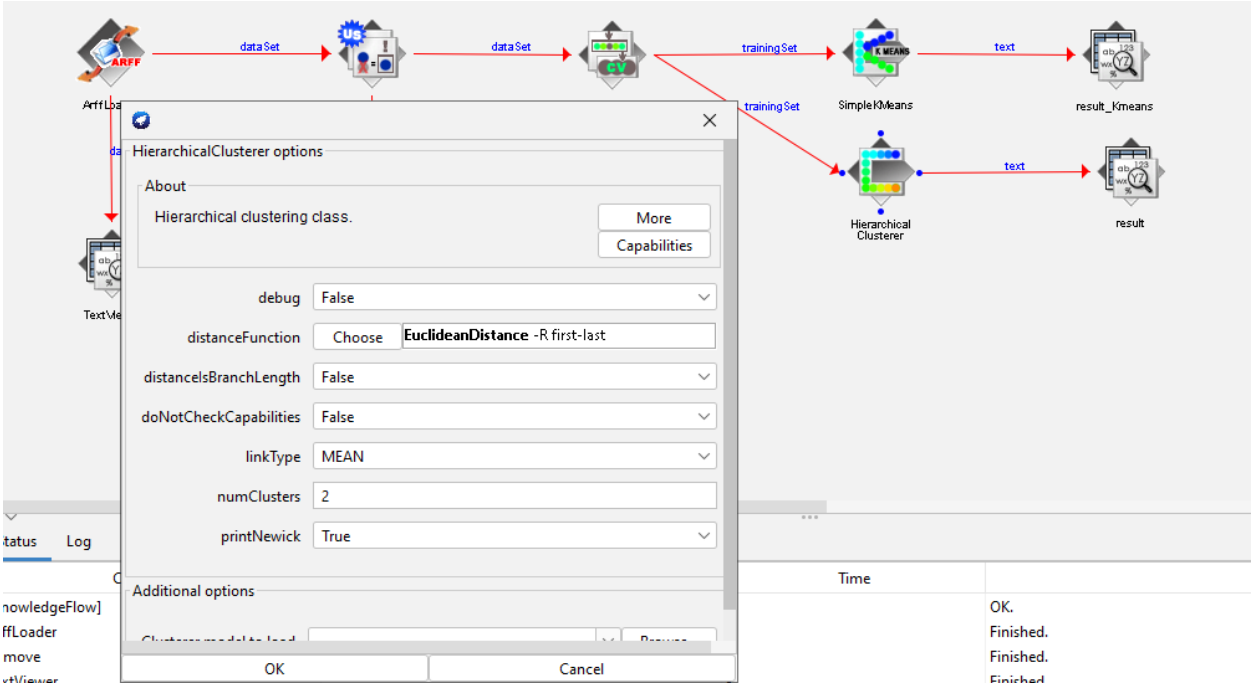
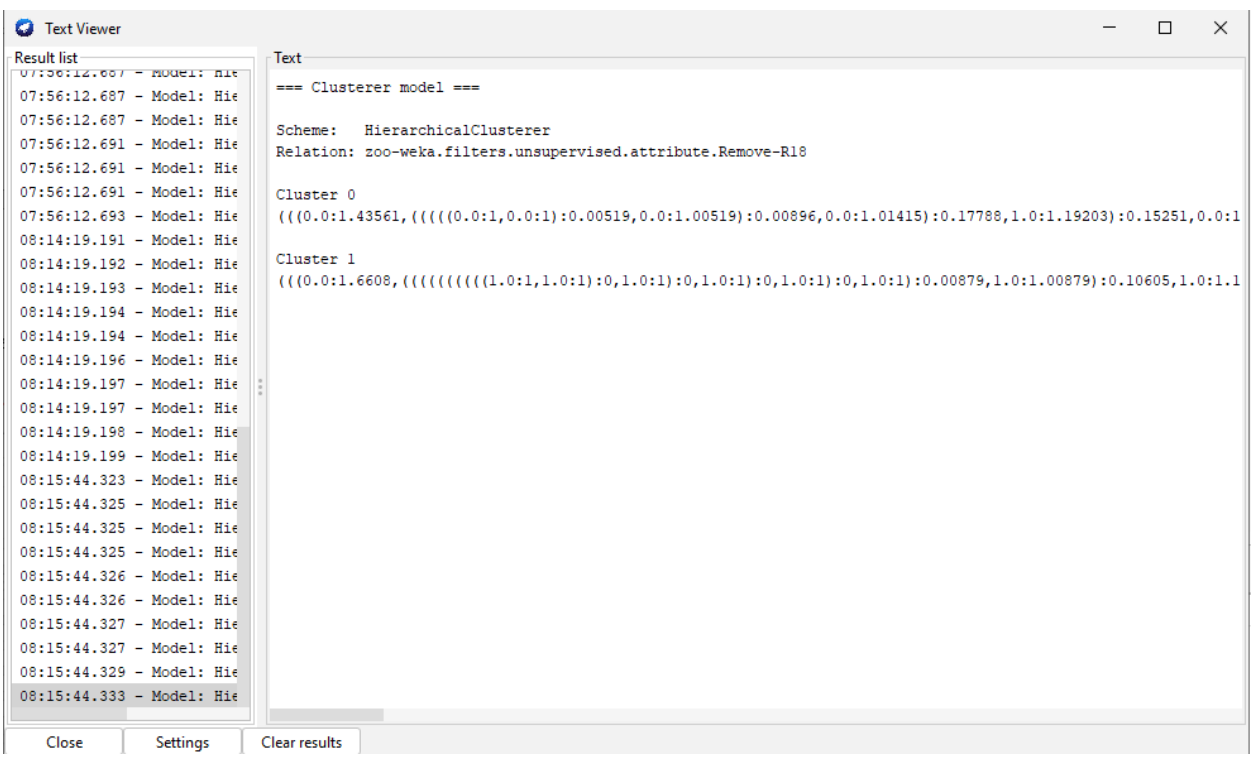

HierarchicalClusterer options
About
Hierarchical clustering class.
More
Capabilities
debug
False
distanceFunction
Choose
EuclideanDistance -R first-last
distanceIsBranchLength
False
doNotCheckCapabilities
False
linkType
AVERAGE
numClusters
2
printNewick
True
Additional options
Clusterer model to load
Browse...
OK
Cancel
result_Kmeans
text
result
OK.
Finished.
Finished.

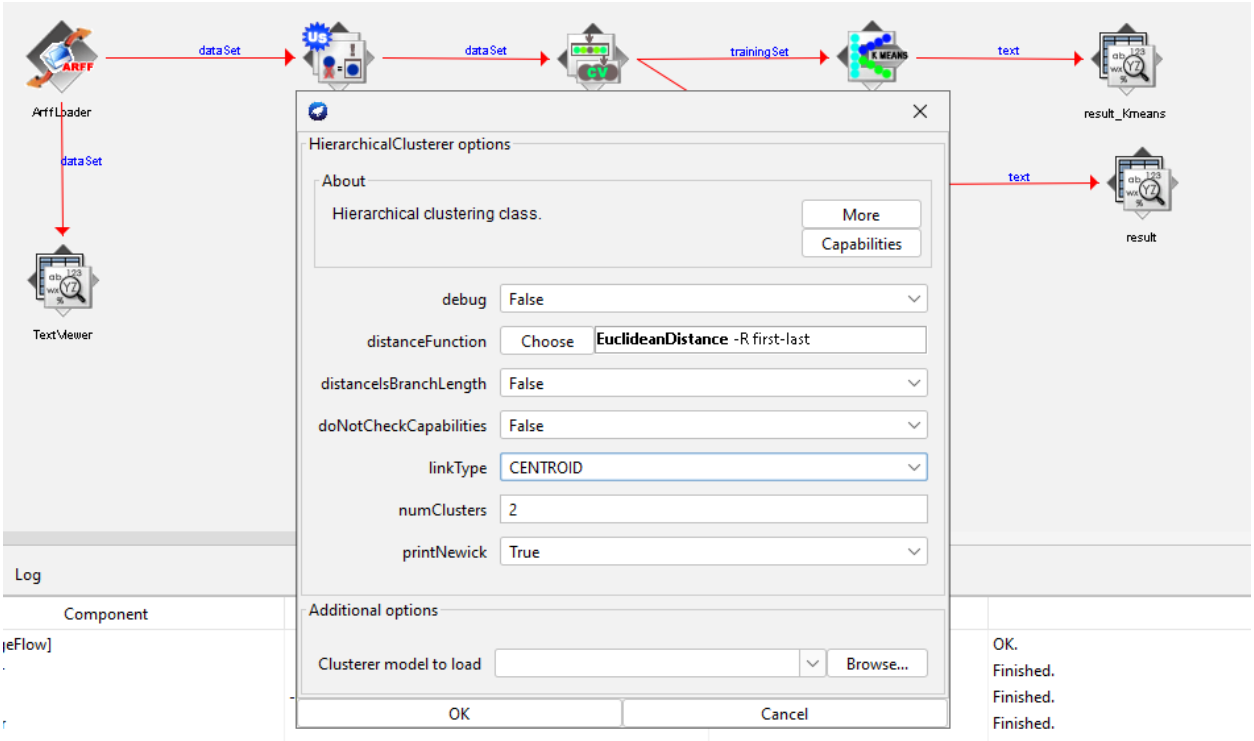
Text Viewer
Result list
07:54:19.304 - Model: Hie
07:54:19.305 - Model: Hie
07:54:19.305 - Model: Hie
07:54:19.306 - Model: Hie
07:54:19.307 - Model: Hie
07:54:19.308 - Model: Hie
07:56:12.683 - Model: Hie
07:56:12.684 - Model: Hie
07:56:12.687 - Model: Hie
07:56:12.687 - Model: Hie
07:56:12.687 - Model: Hie
07:56:12.687 - Model: Hie
07:56:12.691 - Model: Hie
07:56:12.691 - Model: Hie
07:56:12.691 - Model: Hie
07:56:12.693 - Model: Hie
08:14:19.191 - Model: Hie
08:14:19.192 - Model: Hie
08:14:19.193 - Model: Hie
08:14:19.194 - Model: Hie
08:14:19.194 - Model: Hie
08:14:19.196 - Model: Hie
08:14:19.197 - Model: Hie
08:14:19.197 - Model: Hie
08:14:19.198 - Model: Hie
08:14:19.199 - Model: Hie
Text
=== Clusterer model ===
Scheme: HierarchicalClusterer
Relation: zoo-weka.filters.unsupervised.attribute.Remove-R18
Cluster 0
((((((0.0:1.64955,((((0.0:1,0.0:1):0.00778,0.0:1.00778):0.01533,0.0:1.02311):0.43573,1.0:1.45884):0.1907):0.0
Cluster 1
((((((((((((1.0:1,1.0:1):0,1.0:1):0,1.0:1):0,1.0:1):0,1.0:1):0.03078,1.0:1.03078):0.40223,1.0:1.43301):0.0865
Close
Settings
Clear results

j. Start using the HierarchicalClusterer with Mean Link.





k. Start using the HierarchicalClusterer with Centroid Link.



Exercise 5: Using Java programming language to cluster data for the above exercises

- Kmeans

```java
package ClusteringData;

import weka.clusterers.HierarchicalClusterer;
import weka.clusterers.SimpleKMeans;
import weka.core.EuclideanDistance;
import weka.core.Instance;
import weka.core.Instances;
import weka.core.ManhattanDistance;
import weka.core.converters.ConverterUtils.DataSource;

public class ClusteringData {
    DataSource src;
    Instances dataset;
    SimpleKMeans km;
    HierarchicalClusterer hc;

    public void loadDataset(String filename) throws Exception {
        src = new DataSource(filename);
        dataset = src.getDataSet();

        dataset.setClassIndex(-1);
    }

    public void buildClusterKmeans(int k , boolean isManhattan) throws Exception {
        km = new SimpleKMeans();
        km.setNumClusters(k);
        if(isManhattan)
            km.setDistanceFunction(new ManhattanDistance());
        else
            km.setDistanceFunction(new EuclideanDistance());

        km.buildClusterer(dataset);
    }

    public void assignNumCluster() throws Exception {
        for (Instance inst: dataset) {
            int numcluster = km.clusterInstance(inst);
            System.out.println(inst.toString()+" thuộc cụm: "+numcluster);
        }
    }
}
```

```java
public class Demo {
    public static void main(String[] args) {
        try {
            ClusteringData cd = new ClusteringData();

            // Load dữ liệu Zoo
            cd.loadDataset("T:\\zoo.arff");

            // Chạy KMeans với 7 cụm
            cd.buildClusterKmeans(7, true); // true = Manhattan

            // In kết quả phân cụm
            cd.assignNumCluster();

        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

```
<terminated> Demo [Java Application] C:\Java\jdk-17\bin\javaw.exe (Mar 31, 2026, 8:40:40 AM – 8:40:40 AM) [pid: 8636]
aardvark,true,false,false,true,false,false,true,true,true,true,false,false,4,false,false,true,mammal thuộc cụm: 5
antelope,true,false,false,true,false,false,false,true,true,true,false,false,4,true,false,true,mammal thuộc cụm: 5
bass,false,false,true,false,false,true,true,true,true,false,false,true,0,true,false,false,fish thuộc cụm: 3
bear,true,false,false,true,false,false,true,true,true,true,false,false,4,false,false,true,mammal thuộc cụm: 5
boar,true,false,false,true,false,false,true,true,true,true,false,false,4,true,false,true,mammal thuộc cụm: 5
buffalo,true,false,false,true,false,false,false,true,true,true,false,false,4,true,false,true,mammal thuộc cụm: 5
calf,true,false,false,true,false,false,false,true,true,true,false,false,4,true,true,true,mammal thuộc cụm: 5
carp,false,false,true,false,false,true,false,true,true,false,false,true,0,true,true,false,fish thuộc cụm: 3
catfish,false,false,true,false,false,true,true,true,true,false,false,true,0,true,false,false,fish thuộc cụm: 3
cavy,true,false,false,true,false,false,false,true,true,true,false,false,4,false,true,false,mammal thuộc cụm: 5
cheetah,true,false,false,true,false,false,true,true,true,true,false,false,4,true,false,true,mammal thuộc cụm: 5
chicken,false,true,true,false,true,false,false,false,true,true,false,false,2,true,true,false,bird thuộc cụm: 6
chub,false,false,true,false,false,true,true,true,true,false,false,true,0,true,false,false,fish thuộc cụm: 3
clam,false,false,true,false,false,false,true,false,false,false,false,false,0,false,false,false,invertebrate thuộc cụm: 1
crab,false,false,true,false,false,true,true,false,false,false,false,false,4,false,false,false,invertebrate thuộc cụm: 1
crayfish,false,false,true,false,false,true,true,true,false,false,false,false,false,6,false,false,false,invertebrate thuộc cụm: 1
crow,false,true,true,false,true,false,true,false,true,true,false,false,2,true,false,false,bird thuộc cụm: 2
deer,true,false,false,true,false,false,false,true,true,true,false,false,4,true,false,true,mammal thuộc cụm: 5
dogfish,false,false,true,false,false,true,true,true,true,true,false,false,true,0,true,false,true,fish thuộc cụm: 3
dolphin,false,false,false,true,false,true,true,true,true,true,false,true,0,true,false,true,mammal thuộc cụm: 0
dove,false,true,true,false,true,false,false,false,true,true,false,false,2,true,true,false,bird thuộc cụm: 6
duck,false,true,true,false,true,true,false,false,true,true,false,false,2,true,false,false,bird thuộc cụm: 2
elephant,true,false,false,true,false,false,false,false,true,true,true,false,false,4,true,false,true,mammal thuộc cụm: 5
flamingo,false,true,true,false,true,false,false,false,true,true,false,false,2,true,false,true,bird thuộc cụm: 6
```

- **HierarchicalClusterer with Single Link**

```java
    }

    public void buildHierarchicalSingleLink(boolean isManhattan, String type) throws Exception {
        hc = new HierarchicalClusterer();
        hc.setLinkType(new SelectedTag(type, HierarchicalClusterer.TAGS_LINK_TYPE));
        hc.buildClusterer(dataset);
    }

    // In kết quả phân cụm cho HierarchicalClusterer
    public void assignHierarchicalCluster() throws Exception {
        for (int i = 0; i < dataset.numInstances(); i++) {
            Instance inst = dataset.instance(i);
            int clusterNum = hc.clusterInstance(inst);
            System.out.println(inst + " -> Cluster: " + clusterNum);
        }
    }
}
```

```java
            System.out.println("=============SINGLE LINK================");
            cd.buildHierarchicalSingleLink(false, "SINGLE"); // có thể "COMPLETE", "AVERAGE", "CENTROID"
            cd.assignHierarchicalCluster();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Problems Servers Terminal Data Source Explorer Properties Console

```
<terminated> Demo [Java Application] C:\Java\jdk-17\bin\javaw.exe (Mar 31, 2026, 8:50:20 AM – 8:50:20 AM) [pid: 5992]
worm,false,false,true,false,false,false,false,false,false,true,false,false,0,false,false,false,invertebrate thuộc cụm: 1
wren,false,true,true,false,true,false,false,false,true,true,false,false,2,true,false,false,bird thuộc cụm: 6
=============SINGLE LINK================
aardvark,true,false,false,true,false,false,true,true,true,true,false,false,4,false,false,true,mammal -> Cluster: 0
antelope,true,false,false,true,false,false,false,true,true,true,false,false,4,true,false,true,mammal -> Cluster: 0
bass,false,false,true,false,false,true,true,true,true,false,false,true,0,true,false,false,fish -> Cluster: 0
bear,true,false,false,true,false,false,true,true,true,true,false,false,4,false,false,true,mammal -> Cluster: 0
boar,true,false,false,true,false,false,true,true,true,true,false,false,4,true,false,true,mammal -> Cluster: 0
buffalo,true,false,false,true,false,false,false,true,true,true,false,false,4,true,false,true,mammal -> Cluster: 0
calf,true,false,false,true,false,false,false,true,true,true,false,false,4,true,true,true,mammal -> Cluster: 0
carp,false,false,true,false,false,true,false,true,true,false,false,true,0,true,true,false,fish -> Cluster: 0
catfish,false,false,true,false,false,true,true,true,true,false,false,true,0,true,false,false,fish -> Cluster: 0
cavy,true,false,false,true,false,false,false,true,true,true,false,false,4,false,true,false,mammal -> Cluster: 0
cheetah,true,false,false,true,false,false,true,true,true,true,false,false,4,true,false,true,mammal -> Cluster: 0
chicken,false,true,true,false,true,false,false,false,true,true,false,false,2,true,true,false,bird -> Cluster: 0
chub,false,false,true,false,false,true,true,true,true,false,false,true,0,true,false,false,fish -> Cluster: 0
clam,false,false,true,false,false,false,true,false,false,false,false,false,0,false,false,false,invertebrate -> Cluster: 0
crab,false,false,true,false,false,true,true,false,false,false,false,false,4,false,false,false,invertebrate -> Cluster: 0
crayfish,false,false,true,false,false,true,true,false,false,false,false,false,6,false,false,false,invertebrate -> Cluster: 0
crow,false,true,true,false,true,false,true,false,true,true,false,false,2,true,false,false,bird -> Cluster: 0
deer,true,false,false,true,false,false,false,true,true,true,false,false,4,true,false,true,mammal -> Cluster: 0
dogfish,false,false,true,false,false,true,true,true,true,false,false,true,0,true,false,true,fish -> Cluster: 0
dolphin,false,false,false,true,false,true,true,true,true,true,false,true,0,true,false,true,mammal -> Cluster: 0
dove,false,true,true,false,true,false,false,false,true,true,false,false,2,true,true,false,bird -> Cluster: 0
```